

DATABASE MANAGEMENT SYSTEMS

II B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS05	PCC	L	T	P	C	CIE	SEE	Total
		3	-	-	3	30	70	100

COURSE OVERVIEW:

This course introduces the core principles and techniques required in the design and implementation of database systems. This introductory application-oriented course covers the relational database systems RDBMS - the predominant system for business, scientific and engineering applications at present. It includes Entity-Relational model, Normalization, Relational model, Relational algebra, and data access queries as well as an introduction to SQL. It also covers essential DBMS concepts such as: Transaction Processing, Concurrency Control and Recovery. It also provides students with theoretical knowledge and practical skills in the use of databases and database management systems in information technology applications.

COURSE OBJECTIVES:

The course should enable the students to:

1. **Discuss** the basic database concepts, applications, data models, schemas and instances.
2. **Design** Entity Relationship model for a database.
3. **Demonstrate** the use of constraints and relational algebra operations.
4. **Describe** the basics of SQL and construct queries using SQL
5. **Understand** the importance of normalization in databases.
6. **Demonstrate** the basic concepts of transaction processing and concurrency control.
7. **Understand** the concepts of database storage structures and identify the access techniques.

COURSE OUTCOMES:

At the end of the course the students are able to:

1. Use the basic concepts of Database Systems in Database design
2. Apply SQL queries to interact with Database
3. Apply normalization on database design to eliminate anomalies
4. Analyze database transactions and can control them by applying ACID properties
5. Analyze physical database storage system of database.

UNIT-I	GROUP OF INSTITUTIONS
Introduction: Database system applications, Database system Vs file systems, Advantage of a DBMS, Describing and storing data in a DBMS, Structure of a DBMS, People who work with databases. Entity Relationship Model (ER Model): Database Design and ER Diagrams, Entities Attributes and Entity sets, Features of ER Model, Conceptual design with the ER model.	
UNIT-II	
Introduction to relational model: Structure of Relational Databases, Database Schema, Keys, Schema Diagrams, Relational Query Languages, Relational Operations. Introduction to SQL: Overview of the SQL Query Language, SQL Data Definition language, Basic Structure of SQL Queries, Basic operations, Set Operations, NULL Values, Aggregate Functions, Nested Sub Queries, JOIN Expressions, Views, Transactions, Integrity Constraints, SQL Data types and Schemas, Functions and Procedures, Triggers.	
UNIT-III	
Relational Algebra and Calculus: Relational Algebra, Tuple Relational Calculus, Domain Relational Calculus. Schema Refinement and Normal Forms: Introduction to schema refinement, Functional Dependencies, Reasoning about FDs, Normal Forms: 1NF, 2NF, 3NF, Boyce Codd Normal Form, Properties of decompositions, Multi valued Dependencies, Fourth Normal Form, Join Dependencies and Fifth Normal Form.	

UNIT-IV		
Transaction Management: Transaction Concept, A simple transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Serializability. Concurrency Control and Recovery System: Lock based protocols, Deadlock handling, Multiple granularity, Time stamp based protocols, Validation based protocols. Failure Classification, Storage, Recovery and Atomicity, Failure with Non-volatile Storage, Remote backup systems.		
UNIT-V		
Storage and File Structure: Overview of Physical Storage Media, Magnetic Disk and Flash Storage, RAID, Tertiary Storage, File Organization, Organization of Records in Files, Data Dictionary storage. Indexing and Hashing: Basic Concepts, Ordered Indices, B+ Tree Index Files, Multiple Key access, Static Hashing, Dynamic Hashing, Comparison of Ordered Indexing and Hashing, Bitmap Indices.		
TEXT BOOKS:		
1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Sixth Edition, Tata McGraw Hill, 2011. 2. Raghurama Krishnan, Johannes Gehrke, "Data base Management Systems", TATA McGraw Hill, 3rd Edition, 2007. 3. R.P. Mahapatra & Govind Verma, Database Management Systems, Khanna Publishing House, 2013.		
REFERENCE BOOKS:		
1. Peter Rob, Carlos Coronel, Database Systems Design Implementation and Management, 7th edition, 2009. 2. Scott Urman, Michael McLaughlin, Ron Hardman, "Oracle database 10g PL/SQL programming ", 6th edition, Tata McGraw Hill, 2010 3. .K.Singh, "Database Systems Concepts, Design and Applications", First edition, Pearson Education, 2006. 4. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Fourth Edition, Pearson / Addison wesley, 2007.		
WEB REFERENCES:		
1. http://www.learnadb.com/databases/how-to-convert-er-diagram-to-relational-database 2. https://www.w3schools.com/sql/sql_create_table.asp 3. http://www.edugrabs.com/conversion-of-er-model-to-relational-model/?upm_export=print 4. http://ssyu.im.ncnu.edu.tw/course/CSDB/chap14.pdf 5. http://web.cs.ucdavis.edu/~green/courses/ecs165a-w11/8-query.pdf		
E-TEXT BOOKS:		
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MOOC Course		
1. https://www.mooc-list.com/tags/dbms-extensions 2. https://onlinecourses.nptel.ac.in/noc18_cs15/preview		